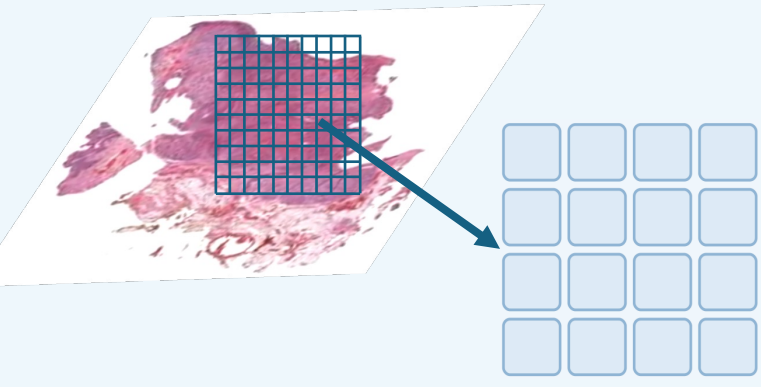


Whole-Slide Image (.svs)



Patch Features ($\{x_i\}_{i=1}^N \in R^d$)
(256x256 @ 10x, pre-extracted)

Frozen CONCH Image Encoder

TISSUE Prompts (8 Fixed)

tumor	stroma
lymphocyte	nercrosis
normal epith.	vessel
adipose	background

Frozen CONCH Text Encoder
(one encoder, shared)

CURRENT DIAGNOSTIC TASK

- class 1: "adenocarcinoma ..."
- class 2: "squamous cell ..."
- ⋮
- class C: "normal tissue ..."

Training & Inference

Training Only ----->
(loss, utility, memory, gradients)

HIERARCHICAL EVIDENCE SELECTION (within a task)

solid path = shared by training AND inference

① Prompt-Guided Grouping

Patch features $\{x_i\}$

Cosine similarity to prompts
(Nx8)

argmax assign $\rightarrow a_i \in \{1..8\}$

Group Members

G_1 G_2 ... G_J mean of member features

Group Prototypes ($J = 8$)

g_1 g_2 ... g_J
 $g_j = \text{mean}(x_i \in G_j)$

② Group Selector ($F_g + \text{LoRA}$)

MLP (2 layers)
 base frozen **LoRA**

Group score r_j

Budget Allocation
largest-remainder $\sum_j b_j = B$

Group Quota b_j 3 2 2 1 0 ... 0

Patch-wise Semantic Prior

$$\mathcal{L}_{sem} = \text{KL}(\text{prior} \parallel \text{score})$$

③ Patch Selector ($F_p + \text{LoRA}$)

x_i in G_j (group members)

MLP (2 layers)
 base frozen **LoRA**

Patch Score s_i

Top b_j within each group

Selected Evidence $\mathcal{P}$ ($|\mathcal{P}| = B$)



④ Frozen Diagnosis Head

Score-weighted pooling + L2 norm

Cosine with class text embedding $\{t_c\}$

Prediction $p(y | \mathcal{P})$

Inference Ends Here

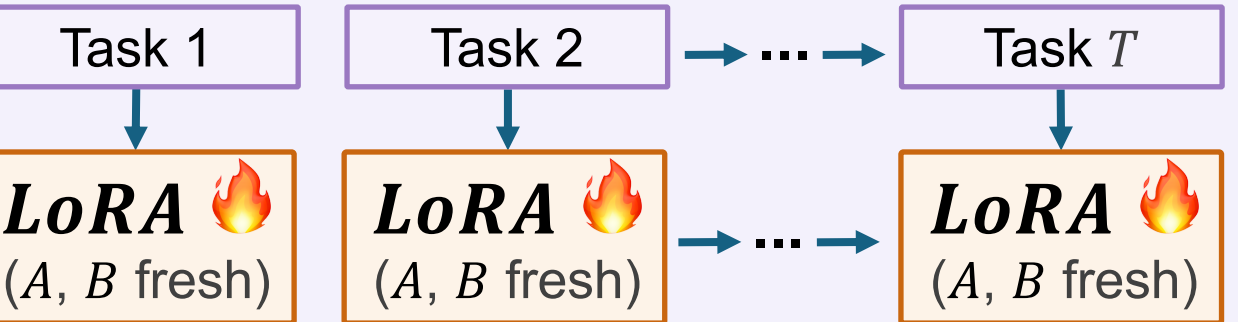
Diagnostic Utility

$$\mathcal{L}_{diag} = \text{CE}(\text{logits}, y)$$

u : conuterfactual gain

$$\mathcal{L}_{util} = \text{KL}(\text{softmax } u \parallel \text{score})$$

CONTINUAL LEARNING ACROSS TASKS



Merge into Shared Selector

write ΔW into BOTH selector MLPs (F_g and F_p)

One Updated Shared Selector

= the same 2 MLPs (F_g and F_p) drawn in ② ③

Selection Memory $\mathcal{M}$ ($|\mathcal{M}| \leq 512$)

(no patch/feature stored – reloaded by *slide_{id}*)

• Group Score
 r_{old}

• Patch Score
 s_{old}

• Utility
 u_{old}

• Replay indices
 $slide_{id} + \text{candidate}$

$\mathcal{L}_{KD}$

$$\mathcal{L}_{KD} = \text{KL}(r_{old} \parallel r_{new}) + \text{KL}(s_{old} \parallel s_{new})$$

group & patch-level KD

$\mathcal{L}_{eq}$

$$\max(0, U_{old} - U_{new})$$

$\mathcal{L}_{replay}$

$$\mathcal{L}_{replay} = \mathcal{L}_{diag} \text{ on replayed samples}$$

Total Loss

$$\mathcal{L}_{evidence} = \mathcal{L}_{diag} + \beta_s \mathcal{L}_{sem} + \beta_u \mathcal{L}_{util}$$

$$\mathcal{L}_{total} = \mathcal{L}_{evidence} + \lambda_{KD} \mathcal{L}_{KD} + \lambda_{eq} \mathcal{L}_{eq} + \lambda_r \mathcal{L}_{replay}$$